Retail Sales Analytics

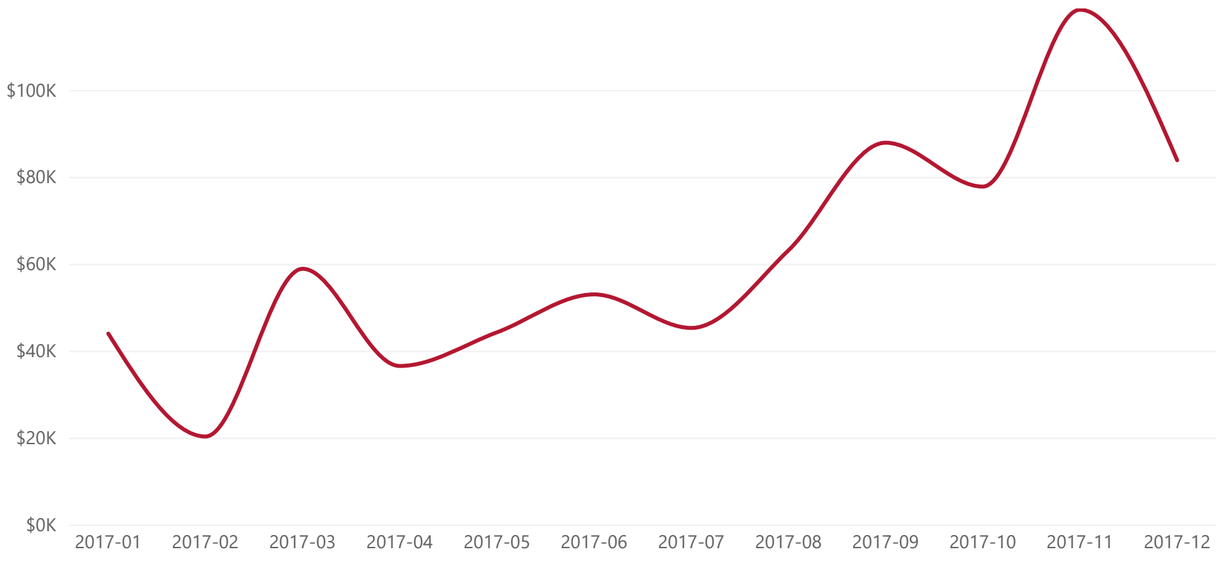
Total Orders
1,687

Profit Margin %
12.74%

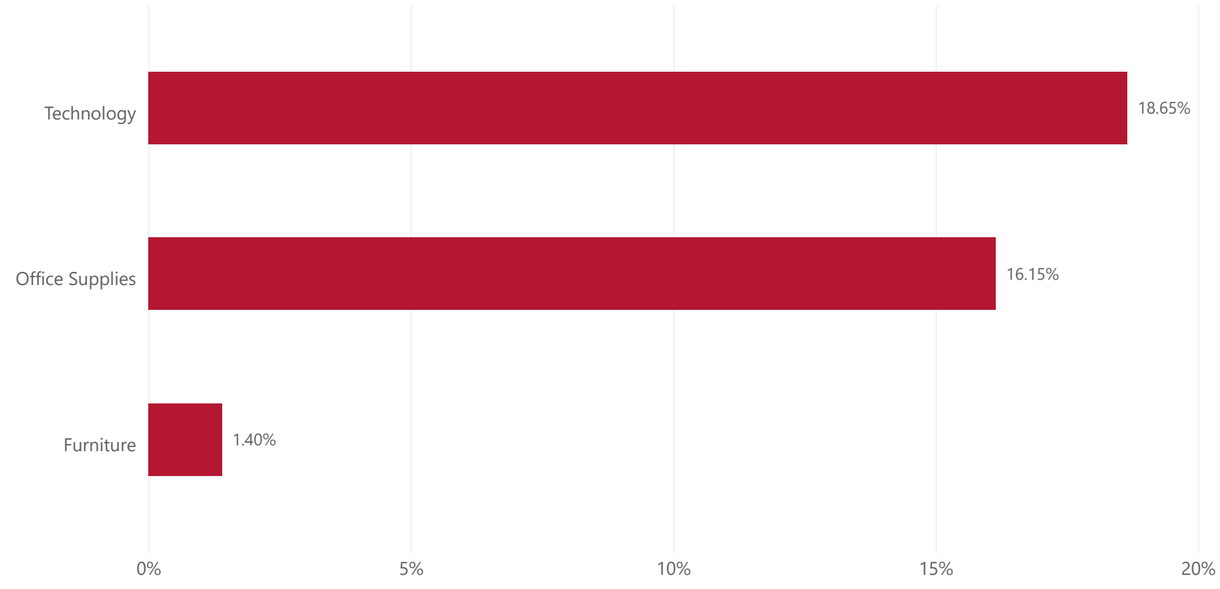
Total Profit
$93.44K

Total Sales
$733.22K

Total Sales
by Year Month



Profit Margin by Category



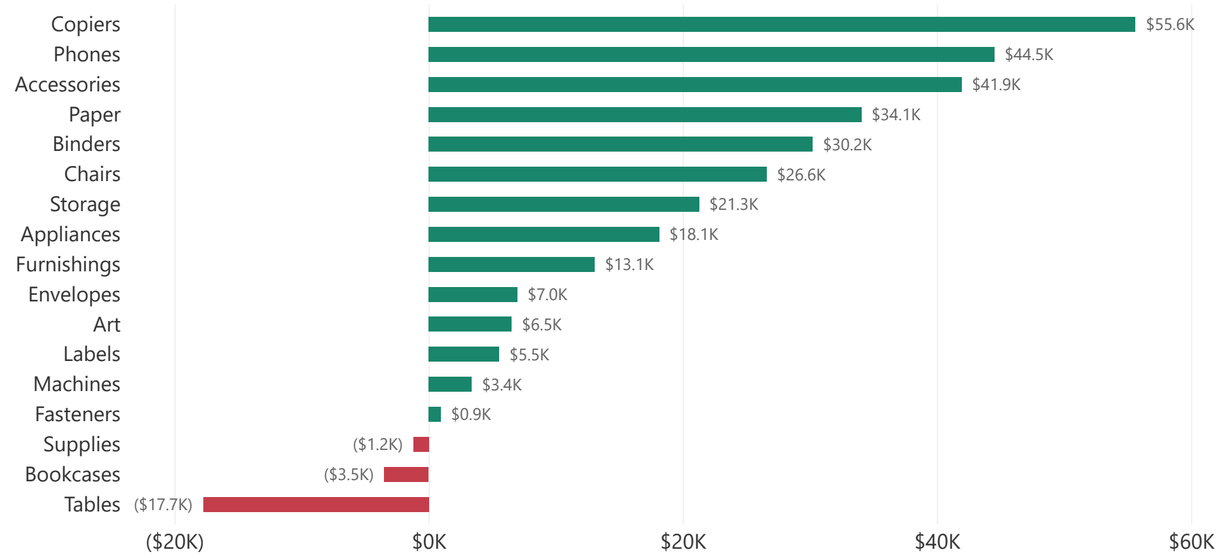
Year



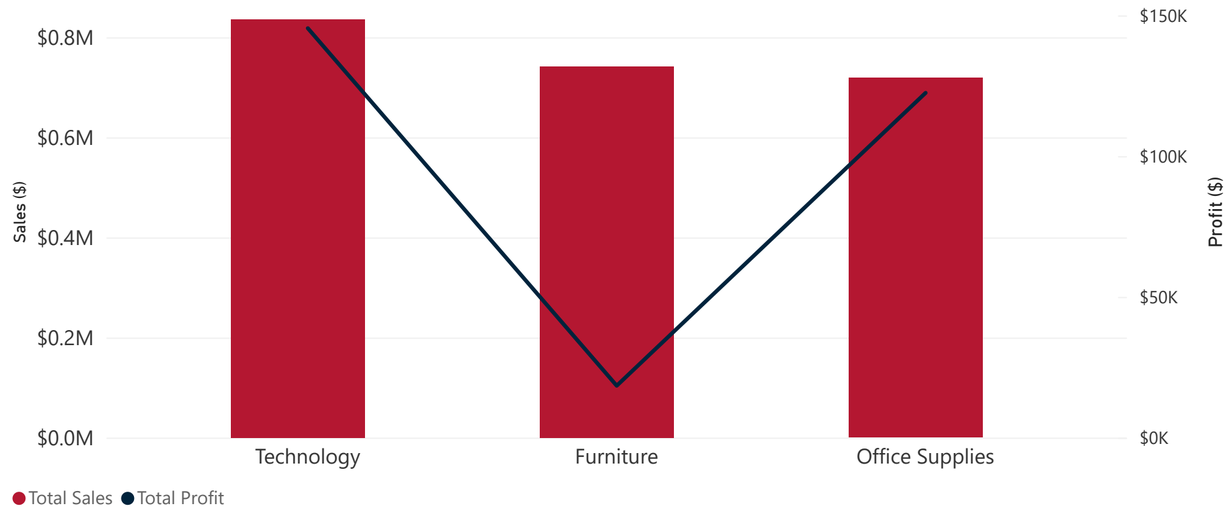
Sales YoY KPI
20.36%

Sales & Profitability Analysis

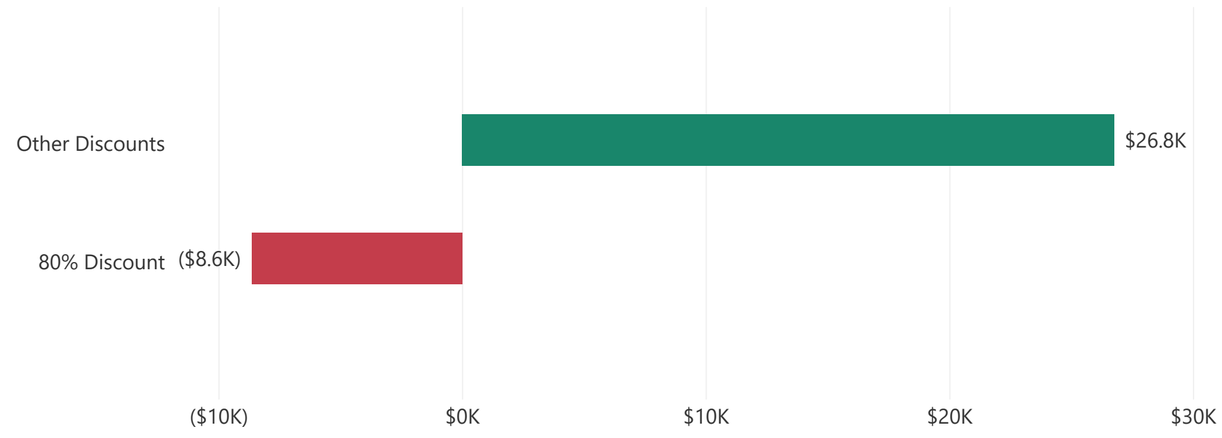
Total Profit
by Sub-Category



Sales vs Profit by Category



Appliances Profit by Discount Group



Yearly Sales Performance

Year	Total Sales	Previous Year Sales	Sales YoY Growth %	Total Profit	Profit Margin %
2014	$484,247.50			$49,543.97	10.23%
2015	$470,532.51	$484,247.50	-2.83%	$61,618.60	13.10%
2016	$609,205.60	$470,532.51	29.47%	$81,795.17	13.43%
2017	$733,215.26	$609,205.60	20.36%	$93,439.27	12.74%
Total	**$2,297,200.86**			**$286,397.02**	**12.47%**

Statistical Insights

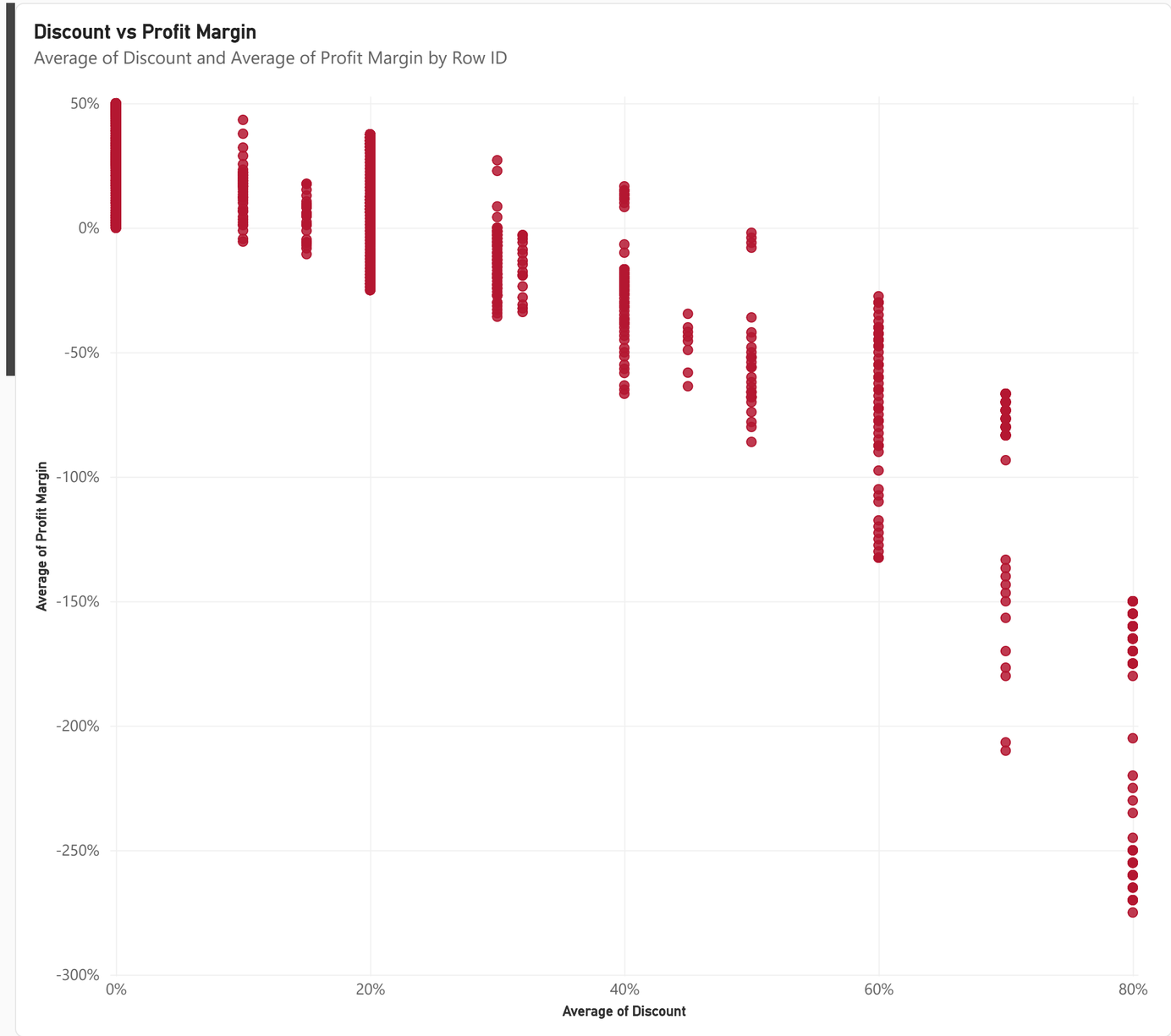
Pearson Correlation (r) = −0.8645

Regression R^2 = 0.7507

Discount Coefficient = −1.9515

Observations = 9,994

+10 pp Discount → −19.52 pp Margin



Key Statistical Findings

1- Strong Negative Correlation

Higher discounts are strongly associated with lower profit margins (r = −0.8645).

2- Category Differences

The relationship between discounts and profit margins varies significantly across product categories.

3- Model Limitations

Extreme observations and non-normal residuals warrant caution.
Statistical association does not establish causation.